

**Report**  
**Of**  
**SUMMER INTERNSHIP REPORT ON SOFTWARE**  
**ENGINEERING**

**(Backend Development with Java & Spring Boot)**  
**at Code-B Solutions Pvt. Ltd.**

Submitted

To

**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

**B.Tech CSE**



By

**Roll Number of Student: CS-2341816**

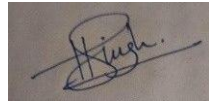
**Name of Student: HARJIT SINGH BHADAURIYA**

**Batch: 2026-27**

## **CANDIDATE'S DECLARATION**

I, Harjit Singh Bhadauriya, do hereby declare that the internship report titled “Summer Internship Report on Software Engineering (Backend Development with Java & Spring Boot) at Code-B Solutions Pvt. Ltd.” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

A handwritten signature in dark ink, appearing to read 'Harjit', is written over a light-colored rectangular background.

Student Name: Harjit Singh Bhadauriya  
Roll No.: CS-2341816

## ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to express my sincere gratitude to Code-B Solutions Pvt. Ltd. for providing me the opportunity to gain practical software engineering experience as a Software Engineering Intern, and to IILM University, Greater Noida for providing the academic platform and guidance for this internship evaluation.

I express my gratitude to my parents for their blessings.

Signature:

A handwritten signature in blue ink, appearing to read 'Harjit Singh', is written over a light blue horizontal line.

Student Name: Harjit Singh Bhadauriya

Roll No.: CS-2341816

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# INTERNSHIP COMPLETION CERTIFICATE



CIN : U62013MH2023PTC399457

Code B Solutions Pvt Ltd, Bungalow No-23, RDP 2, MHADA Colony, Mulund East, Mumbai 400081

Dear Harjit Singh Bhadauriya,

We are pleased to formally acknowledge the successful completion of your internship as an **Software Engineering Intern** at CODE-B SOLUTIONS PRIVATE LIMITED. Your dedication and contributions from 09-03-2026 to 01-08-2026 have been greatly appreciated.

During your time with us, you demonstrated a strong understanding of Java and its Framework Spring Boot and also a commitment towards learning and collaboration. Your efforts have contributed positively to our projects, and we commend your growth throughout this experience. Sincerely,

A handwritten signature in black ink, appearing to be 'Raj', is written over a faint, light blue circular stamp.

Code-B Solutions Pvt Ltd.

Internship Certification generated on 01-08-2026

## **4. Project Description**

### **4.1 Introduction**

This report documents the Summer Internship undertaken as part of the B.Tech Computer Science and Engineering programme (7th Semester, Batch 2026–27) at IILM University, Greater Noida. The internship was carried out with Code-B Solutions Pvt. Ltd., Mumbai, in the role of Software Engineering Intern, from 09 March 2026 to 01 August 2026 (a duration of approximately six months).

The internship was designed to bridge academic concepts with real-world software engineering practices, providing hands-on exposure to Java backend development, REST API engineering, database performance tuning, cloud deployment, DevOps automation, and production-oriented testing practices. This report summarises the organisation, the problem addressed, the objectives, scope, tools used, system architecture, methodology followed, and the outcomes achieved during the internship.

### **4.2 Organization Profile**

Code-B Solutions Pvt. Ltd. is a software solutions company headquartered at Bungalow No-23, RDP 2, MHADA Colony, Mulund East, Mumbai – 400081 (CIN: U62013MH2023PTC399457). The internship was carried out remotely under the organisation's engineering team, which focuses on building scalable backend systems, REST APIs, and cloud-deployed services for its clients.

### **4.3 Problem Statement**

Modern backend systems are expected to serve high volumes of requests reliably while remaining maintainable, secure, and easy to scale. Many existing services face challenges such as inconsistent API response contracts, unoptimised database queries leading to high latency, manual and error-prone deployment processes, and insufficient automated testing, all of which increase the risk of production failures. The internship addressed the problem of designing, building, and deploying production-grade REST APIs and backend services that are reliable, performant, and supported by automated CI/CD and testing pipelines.

### **4.4 Project Objectives**

- To design and develop production-grade REST APIs using Java and Spring Boot.

- To implement centralized exception handling and input validation for reliable client integration.
- To analyse and optimize PostgreSQL database schemas, queries, and indexing strategies.
- To containerize backend services using Docker and deploy them on AWS EC2.
- To build automated CI/CD pipelines using GitHub Actions to reduce manual intervention during releases.
- To develop unit, integration, and API contract tests to improve backend reliability.
- To apply clean engineering principles and Domain-Driven Design concepts across the codebase.

#### 4.5 Scope of the Project

The scope of the internship spanned the complete backend development lifecycle — from API design and implementation to database optimization, containerization, CI/CD automation, cloud deployment, and quality assurance. The work covered:

- Backend & REST API Engineering – designing and deploying 15+ production-grade REST APIs.
- Database Optimization – improving PostgreSQL schema design, indexing, and query execution plans.
- Cloud & DevOps – Docker-based containerization, GitHub Actions CI/CD, and AWS EC2 deployment.
- Quality & Reliability Engineering – unit/integration testing with JUnit and API contract testing.

#### 4.6 Technologies and Tools Used

| Category                   | Technologies / Tools                                              |
|----------------------------|-------------------------------------------------------------------|
| Backend                    | Java, Spring Boot, Spring Security, Spring Data JPA, RESTful APIs |
| Database & Cloud           | PostgreSQL, AWS EC2                                               |
| DevOps & Development Tools | Docker, GitHub Actions, CI/CD, Git, Maven, Linux                  |
| Testing & API Tools        | JUnit, Integration Testing, API Contract Testing, Postman         |
| Project Management         | JIRA                                                              |

## 4.7 System Architecture

The backend services followed a layered development-to-deployment architecture, where code changes flow from version control through an automated CI/CD pipeline into a containerized, cloud-hosted production environment backed by an optimized relational database. The flow is illustrated below:

| Stage | Component                            |
|-------|--------------------------------------|
| 1     | Developer writes code                |
| 2     | Git Repository (version control)     |
| 3     | GitHub Actions (CI trigger)          |
| 4     | CI/CD Pipeline                       |
| 5     | Docker Container (packaging)         |
| 6     | AWS EC2 (deployment)                 |
| 7     | Spring Boot APIs (application layer) |
| 8     | PostgreSQL (data layer)              |

This architecture ensured that every code change was version-controlled, automatically tested and built, packaged consistently through Docker, and deployed to AWS EC2 with minimal manual intervention, while the Spring Boot application layer interacted with an optimized PostgreSQL database.

## 4.8 Methodology

- Develop – Implement backend features and REST APIs using Java and Spring Boot.
- Version Control – Push code changes to the Git repository.
- Continuous Integration – GitHub Actions automatically triggers the CI/CD pipeline.
- Containerization – Application services are packaged into Docker containers.
- Deployment – Containerized backend services are deployed on AWS EC2.
- Database Layer – Spring Boot services interact with optimized PostgreSQL databases.
- Quality Assurance – Unit, integration, and API contract tests help maintain reliability.

#### 4.9 Expected Outcomes

The internship resulted in the following measurable outcomes:

| Metric                          | Achievement |
|---------------------------------|-------------|
| REST APIs developed             | 15+         |
| Monthly requests handled        | 10K+        |
| Query latency reduction         | ~45%        |
| Backend reliability improvement | ~30%        |
| Internship duration             | 6 months    |

Beyond the quantitative metrics, the internship transformed academic knowledge into practical software engineering experience, providing exposure to the complete backend development lifecycle — from API design and database optimization to testing, containerization, CI/CD, and cloud deployment. It strengthened the understanding of building backend systems that are reliable, scalable, maintainable, and production-oriented, and established a strong foundation for pursuing opportunities in Software Engineering, Java Backend Development, Cloud Engineering, DevOps, and Distributed Systems.

#### 4.10 Certificate of Completion and Other Communication Proof

The Internship Completion Certificate issued by Code-B Solutions Pvt. Ltd., formally acknowledging the successful completion of the Software Engineering Intern role from 09 March 2026 to 01 August 2026 with specific recognition of work in Java and Spring Boot, is attached in the earlier section of this report (Internship Completion Certificate).

## 5. Bibliography/References

The following resources were referred to during the course of the internship and preparation of this report:

- Spring Boot Reference Documentation – <https://docs.spring.io/spring-boot/>
- Spring Security Reference – <https://docs.spring.io/spring-security/reference/>
- Spring Data JPA Reference – <https://docs.spring.io/spring-data/jpa/reference/>
- PostgreSQL Official Documentation – <https://www.postgresql.org/docs/>
- Docker Documentation – <https://docs.docker.com/>
- GitHub Actions Documentation – <https://docs.github.com/actions>
- Amazon Web Services (AWS EC2) Documentation – <https://docs.aws.amazon.com/ec2/>
- JUnit 5 User Guide – <https://junit.org/junit5/docs/current/user-guide/>
- Postman Learning Center – <https://learning.postman.com/>
- Code-B Solutions Pvt. Ltd., Internship Completion Certificate, issued 01-08-2026.